

HPS CUSTOM PRINT & DESIGN

Final Project Implementation Plan

Professional UI/UX + Custom Design Studio + E-Commerce + Order Management + Production + Delivery

Project objective: transform the current single-page HTML prototype into a scalable, secure, production-ready custom printing platform for Bangladesh, with room for future expansion.

Initial prototype baseline

- Products currently modeled: T-Shirt █350, Hoodie █750, Denim █1,200, Bag █300, Cap █250.
- Categories: Men, Women, Unisex.
- Design Studio is based on Fabric.js 5.3.1 with upload, text, duplicate, delete, color, opacity, templates and front/back design concepts.
- Sizes: S, M, L, XL, XXL (+█50), 3XL (+█80).
- Express delivery surcharge: +█100.
- Front print +█100; back print +█100.
- Cart, █100 delivery charge, checkout, COD/bKash/Nagad/Card options and a basic order-status flow already exist in the prototype.
- The current prototype has no real backend/database/payment gateway; cart/order state is in browser memory and disappears after refresh.

Implementation principle: design the user experience and business rules first, then implement persistent data, APIs, storage, payment verification, production workflows and security.

1. Executive Summary

HPS should be implemented as a modular web application rather than continuing to grow the current single HTML file. The final system should separate the storefront, design editor, customer account, checkout, administration and production operations while sharing a single secure backend and database.

Recommended high-level flow:

Customer → Shop/Product → Customize in Design Studio → Design Validation → Cart → Checkout → Payment → Order → Design Review → Printing → Quality Check → Packaging → Shipment → Delivery → Review

The architecture should support both ready-made products and custom designs. Product price, size surcharge, print-side surcharge, express charge, delivery fee, coupon and payment status must be calculated and verified on the server—not trusted from browser JavaScript.

2. Final System Scope

Area	Final capability	Priority
Brand/UI	Responsive visual system, navigation, typography, components, accessibility, mobile UX	P0
Storefront	Homepage, shop, categories, search, filters, product detail, reviews	P0
Design Studio	Fabric.js editor, front/back, templates, uploads, text, image controls, undo/redo, preview	P0
Print Quality	DPI, dimensions, safe area, bleed, transparency and low-resolution warnings	P0
Cart/Checkout	Persistent cart, address, delivery, payment, review and confirmation	P0
Customer Account	Orders, designs, addresses, wishlist, profile, notifications	P1
Admin	Orders, products, customers, designs, templates, inventory, payments, reports	P0
Production	Design approval, print queue, QC, packaging and shipment handoff	P0
Payments	COD, bKash, Nagad and card integration with server-side verification	P0
Delivery	Delivery fee, shipment record, tracking, status updates	P1
Security	Authentication, authorization, upload security, validation, audit logs	P0
Analytics	Sales, revenue, conversion, popular products, abandoned carts, repeat customers	P2

3. Professional UI/UX Implementation

3.1 Brand and Design System

- Create an HPS logo system, favicon, social/share image and basic brand guidelines.
- Define primary/secondary colors, semantic colors, typography scale, button hierarchy, input styles, cards, badges, alerts, modals and notification patterns.
- Use a consistent spacing scale, border radius, shadow levels and icon family.
- Design for mobile first, then tablet and desktop breakpoints.
- Use clear product photography/mockups and avoid relying on placeholder graphics for the production storefront.
- Add loading states, skeletons, empty states, error states and success confirmations for all important actions.

3.2 Main Navigation

- Shop
- Design Studio
- Templates
- Custom Orders
- How It Works
- Search
- Account
- Cart
- Mobile navigation drawer with the same information architecture.

3.3 Homepage

- 1 Hero section with a strong custom-printing message and primary Customize Now CTA.
- 2 Popular products section.
- 3 How It Works: Choose → Design → Order → We Print → We Deliver.
- 4 Design Studio CTA with live design/mockup visual.
- 5 Template categories.
- 6 Why HPS: quality, customization, delivery, support.
- 7 Customer reviews/testimonials.
- 8 FAQ.
- 9 Footer with contact, policies, social links and service information.

3.4 Shop and Product Detail

- Shop filters: category, gender, size, color, price, printing method and availability.
- Product cards: image, name, starting price, rating, availability and Customize button.
- Product detail: front/back/detail images, description, variants, color, size, print options, quantity, delivery estimate and Customize Now.
- Show size/price adjustments clearly before adding to cart.
- Keep product-specific print areas and mockup rules in the database rather than hard-coding them in the browser.

3.5 Checkout UX

Use a clear 5-step checkout: **Customer** → **Delivery** → **Payment** → **Review** → **Confirm**. For Bangladesh, support division/district/area/address and phone-first checkout. Keep the final payable amount visible throughout the process.

4. Design Studio — Core Product Feature

The Design Studio should become the main differentiator of HPS. It must feel like a focused product-customization application rather than a generic canvas demo.

4.1 Editor Layout

- Left toolbar: upload, text, shapes, templates, icons, layers and design assets.
- Center workspace: garment mockup and active print area.
- Right panel: object properties, typography, color, opacity, alignment, position and size.
- Top bar: undo, redo, duplicate, delete, zoom, front/back, preview, save design and add to cart.
- Bottom or side panel: design quality status and print-area information.

4.2 Editing Features

- Upload PNG/JPG/SVG with strict type, size and content validation.
- Text editing with font family, size, weight, alignment, letter spacing and color.
- Move, resize, rotate, crop, duplicate, delete and layer ordering.
- Undo/redo history.
- Front and back canvas state saved independently.
- Templates with one-click loading into the active product.
- Save design to customer account.
- Preview the design on the selected garment mockup.
- Generate a production-ready print file after order confirmation.

4.3 Print-Quality System

- Define exact print area dimensions per product and print side.
- Use a safe area and optional bleed boundary.
- Calculate effective output resolution and display a quality indicator such as Excellent / Good / Low.
- Warn when an image is too small for the selected print dimensions.
- Prefer transparent PNG/SVG where appropriate; preserve transparency.
- Set maximum upload size and reject dangerous or unsupported files.
- Do not let the client define final print dimensions without server-side validation.

4.4 Product Mockup System

Replace the prototype's simple SVG garment concept with real product-specific mockup assets. Start with T-shirt front/back, then add oversized T-shirt, hoodie, polo, cap, tote bag, mug, bottle and sticker as separate product mockup profiles.

5. Database and Data Architecture

Recommended relational database: PostgreSQL or MySQL. PostgreSQL is preferred if the project expects complex reporting, JSON design metadata and future scale.

Core entities

Table	Purpose
users	Customer/admin identity and account data
products	Base product information and pricing
categories	Product/category organization
product_variants	Size, color, stock and variant-specific data
product_images	Gallery/mockup/detail images
print_areas	Front/back print dimensions and safe areas
designs	Saved customer designs and metadata
design_objects	Structured editor object data when required
templates	Reusable design templates
carts / cart_items	Persistent shopping cart
orders / order_items	Order header and purchased items
payments	Gateway/reference/status/verification data
addresses	Customer delivery addresses
shipments	Courier/tracking/status information
production_jobs	Printing and manufacturing workflow
coupons	Discount rules
reviews	Customer product reviews
notifications	Email/SMS/WhatsApp/web notification records
audit_logs	Security/admin action history

Important relationships

- User → many designs, addresses, carts, orders, reviews and notifications.
- Product → many variants, images and print-area definitions.
- Order → many order items, one or more payment records, shipment and production records.
- Order item → references product variant and the selected design/version.
- Design → references the product/print area and stores editor state plus generated assets.

6. Backend and API Architecture

Recommended production stack

Layer	Recommended approach
Frontend	Next.js/React + TypeScript
Editor	Fabric.js, isolated into a reusable Design Studio module
Backend	Node.js with NestJS/Express, or Laravel if PHP is preferred
Database	PostgreSQL or MySQL
Storage	S3-compatible object storage for uploads/design/print files

Layer	Recommended approach
Authentication	Secure session or token-based authentication with role-based authorization
Caching/queues	Redis and background workers when workload grows
Deployment	Linux server/VPS or managed cloud with HTTPS, backups and monitoring

API modules

- Auth API: register, login, logout, password reset, profile.
- Product API: products, categories, variants, images, filters.
- Design API: create, save, update, duplicate, preview, export and delete designs.
- Template API: list, search, categories and template usage.
- Cart API: add, update, remove, merge and validate cart.
- Checkout API: address, shipping calculation, coupon validation, order creation.
- Payment API: payment initiation, callback/webhook, verification and reconciliation.
- Order API: customer order history and status.
- Admin API: products, customers, designs, orders, payments, production and reports.
- Notification API/service: email/SMS/WhatsApp/web notification jobs.

Critical rule: never trust browser-submitted price, discount, shipping fee, payment status or product availability. Recalculate all financial and business rules on the server.

7. Cart, Checkout and Payment Implementation

Cart calculation

- Base product price.
- Variant/size surcharge: XXL +■50; 3XL +■80.
- Print-side surcharge: front +■100; back +■100.
- Express surcharge: +■100.
- Quantity multiplication.
- Coupon/discount rules.
- Delivery fee: current prototype uses ■100; make this configurable by admin and/or delivery zone.
- Final total calculated by backend.

Payment options

- COD.
- bKash.
- Nagad.
- Card payment through a suitable Bangladesh-supported payment gateway.
- Payment callbacks/webhooks must be verified server-side.
- Store gateway transaction/reference identifiers and reconciliation status.
- Never store raw card credentials.

8. Customer Account System

- Profile and phone/email verification.
- Saved addresses.
- Order history and order detail.
- Saved designs.
- Wishlist.
- Payment/order records where appropriate.
- Notification preferences.
- Password/security settings.
- Downloadable invoices/receipts if required.

9. Admin Dashboard

Admin modules

- Dashboard overview: orders, sales, revenue, pending designs, printing queue, shipments.
- Orders: search/filter/status/payment/customer/order details.
- Products: create/edit products, variants, images, print areas and stock.
- Categories and collections.
- Customers.
- Designs and design review.

- Templates.
- Inventory.
- Printing/production.
- Payments.
- Delivery and shipments.
- Coupons.
- Reviews.
- Notifications.
- Reports and analytics.
- Admin users, roles and permissions.
- System settings and audit logs.

Order management screen

- 1 View customer and order details.
- 2 Inspect product, variant, quantity and design.
- 3 Verify payment.
- 4 Approve or reject design with customer-facing note.
- 5 Send approved job to printing.
- 6 Mark quality checked.
- 7 Mark packaged.
- 8 Create shipment/tracking record.
- 9 Mark shipped and delivered.

10. Production Workflow

Recommended state machine:

Order Received → Payment Verified → Design Review → Design Approved → Printing Queue → Printing → Quality Check → Packaging → Shipped → Delivered

- Keep customer-facing status names simple while allowing more detailed internal production states.
- Every status change should create an audit/event record.
- Design rejection must include a reason and allow the customer to revise/resubmit.
- Generate or store production-ready artwork separately from the editable design.
- Keep the original customer design version so production files can be traced to the exact order.

11. Security and Reliability Requirements

The current prototype must not be treated as production-safe until these controls are implemented.

- HTTPS everywhere.
- Secure authentication and password hashing.
- Role-based authorization for customer, staff and administrator accounts.
- CSRF protection where applicable.
- Rate limiting on login, upload, checkout and sensitive APIs.
- Server-side input validation and output encoding.
- Parameterized database queries/ORM protections.
- Secure file-upload handling: MIME validation, extension validation, size limits, randomized storage names, malware scanning where practical, and never execute uploaded files.
- Store uploads outside the executable/public path where possible; serve through controlled URLs.
- Image/SVG sanitization before rendering.
- Do not trust client-side price or payment status.
- Webhook signature verification for payment providers.
- Audit logs for admin actions and financial/order changes.
- Secrets stored in environment/secret management, never in source code.
- Regular database backups and restore testing.
- Error handling that avoids exposing stack traces or internal details.
- Dependency updates and vulnerability scanning before production release.

12. File and Design Storage

Recommended logical storage structure

- product-images/
- product-mockups/
- user-designs/
- design-thumbnails/
- templates/
- print-files/
- order-files/
- invoices/

Store file metadata and references in the database. Use object storage rather than keeping large binary files directly inside the relational database.

13. Notifications

- Order confirmation.
- Payment confirmation.
- Design review requested.
- Design approved.

- Design rejected / revision requested.
- Printing started.
- Quality check completed.
- Order packaged.
- Shipment created.
- Order shipped.
- Order delivered.

Start with email/web notifications; add SMS/WhatsApp integration when operational requirements justify it.

14. Analytics and Business Intelligence

- Orders per day/week/month.
- Revenue and average order value.
- Popular products, sizes and colors.
- Popular templates and design categories.
- Conversion rate.
- Cart abandonment.
- Repeat customers.
- Payment method distribution.
- Production turnaround time.
- Delivery performance.
- Coupon usage and discount impact.

15. Final Development Roadmap

Phase	Implementation	Acceptance result
01	Brand + design system	Consistent HPS visual language and responsive components
02	Homepage	Professional conversion-focused landing page
03	Shop + filters	Products searchable/filterable on mobile and desktop
04	Product details	Variant/print configuration and Customize flow
05	Design Studio	Stable editor with save, front/back, templates and preview
06	Mockup engine	Product-specific front/back mockups and print areas
07	Print quality system	DPI/dimension/safe-area validation and warnings
08	Cart + checkout	Persistent cart and complete checkout
09	Database + backend	Persistent users/products/designs/orders
10	Authentication	Secure customer/admin roles
11	Storage	Secure design and production-file storage
12	Order system	Server-side totals and order lifecycle
13	Admin dashboard	Operational management without editing database manually
14	Payments	Verified COD/bKash/Nagad/card flows
15	Production	Design approval, printing and QC workflow
16	Delivery	Shipment/tracking lifecycle
17	Notifications	Automated customer status updates
18	Security hardening	Upload/API/auth/payment controls validated
19	Testing	Functional, responsive, security and payment tests pass
20	Deployment + monitoring	Production release, backups, logs and monitoring

16. MVP vs Future Expansion

MVP / launch scope

- Homepage, Shop, Product Detail.
- Design Studio with core editing.
- T-shirt/hoodie initial mockups.
- Templates.
- Customer account and saved designs.
- Persistent cart and checkout.
- COD plus at least one verified online payment method.
- Admin orders/products/design review.
- Production workflow through shipment.
- Core security, backups and logging.

Phase 2

- More product types and mockups.
- Advanced template marketplace.

- Wishlist and stronger personalization.
- Courier integration and automated tracking.
- SMS/WhatsApp notifications.
- Coupons and campaigns.
- Advanced analytics.

Phase 3

- AI-assisted design features.
- Customer design marketplace or creator profiles.
- Bulk/B2B orders.
- Multi-vendor support if business model requires it.
- Advanced production planning and inventory forecasting.
- Mobile app/PWA if traffic and operational data justify it.

17. Recommended Project Structure

Example frontend/backend separation:

apps/web — storefront, account, Design Studio UI

apps/admin — administration and production UI

apps/api — authentication, shop, design, cart, checkout, orders, payments

packages/ui — shared design-system components

packages/types — shared TypeScript types

packages/design-engine — Fabric.js editor logic

infra — deployment, database, storage and monitoring configuration

18. Testing and Quality Assurance

- Responsive testing: mobile, tablet and desktop.
- Browser testing: current Chrome/Edge/Firefox/Safari where applicable.
- Design editor testing: upload, text, resize, rotate, undo/redo, front/back, save/load and export.
- Print validation tests with low-resolution, oversized, transparent and unsupported files.
- Cart tests with quantity, size surcharge, print-side surcharge, express and coupons.
- Checkout tests with valid/invalid addresses and payment outcomes.
- Payment webhook and duplicate-callback tests.
- Authorization tests: customer cannot access another customer's orders/designs.
- Admin role/permission tests.
- File upload abuse tests.
- Database backup/restore test.
- Performance tests for image-heavy product pages and design uploads.
- Accessibility checks: keyboard navigation, focus states, labels, contrast and error messages.

19. Production Deployment Checklist

- 1 Configure production domain and HTTPS.
- 2 Configure frontend and backend environments separately.
- 3 Create production database with backups.
- 4 Configure object storage and private upload paths.
- 5 Set secrets/environment variables.
- 6 Configure email/payment/courier integrations.
- 7 Run database migrations.
- 8 Create admin roles and least-privilege permissions.
- 9 Enable logs, monitoring and alerts.
- 10 Run smoke tests and payment sandbox tests.
- 11 Perform security review and dependency audit.
- 12 Create rollback and backup restore procedure.
- 13 Launch with controlled traffic and monitor order/payment flows.

20. Final Acceptance Checklist

Requirement	Done when...
UI/UX	All major screens are consistent, responsive and usable
Shop	Products, variants, filters and details work from database
Design Studio	Designs can be created, saved, loaded, previewed and validated
Print Quality	Low-quality artwork is detected before production
Cart	Cart survives refresh/login and server validates totals
Checkout	Address, shipping, payment and review work end-to-end
Payment	Provider callbacks are verified and payment status is authoritative
Admin	Staff can manage the full order lifecycle
Production	Approved designs flow to printing/QC/packaging
Delivery	Shipment/tracking state is recorded and customer-visible
Security	Auth, authorization, upload, API and secrets controls pass review
Backup	Backup and restore are tested, not merely configured
Launch	Monitoring, logs and rollback plan are active

21. Immediate Next Actions

- 1 **Freeze the current prototype as the UI/feature reference.** Do not keep adding business logic to the single HTML file.
- 2 **Create the design system.** Define HPS colors, typography, spacing, components and responsive rules.
- 3 **Design the full sitemap and screen wireframes.** Homepage, Shop, Product Detail, Design Studio, Cart, Checkout, Account, Admin.
- 4 **Define product/print-area data.** Each product should have variants, images, mockups and front/back print dimensions.
- 5 **Build the Design Studio as a reusable module.** Keep editor state independent from checkout.
- 6 **Build database and API contracts.** Define tables, relationships, validation and server-side price calculation.
- 7 **Implement authentication and persistent designs/cart.**
- 8 **Implement orders and admin workflow.**
- 9 **Integrate payment and delivery providers.**
- 10 **Apply security hardening, testing, backups and monitoring before public launch.**

22. Target Final Architecture

HPS Customer Web → Shop / Design Studio / Templates / Custom Orders → **Cart** → **Checkout** → Payment / Order → **Admin System** → Design Review / Production / Quality / Packaging / Delivery → **Customer Tracking & Reviews**

This plan converts the current prototype into a complete custom-print commerce and production platform while keeping the Design Studio at the center of the customer experience.